

Ontology-based Evaluation

A Two-Week Module for Artificial Intelligence Capstone

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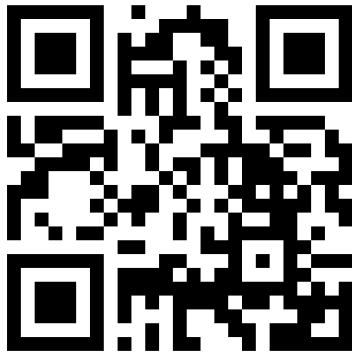
Before We Begin

A quick interaction check with Vevox

- ▶ Please join the Vevox session before we begin.
- ▶ We will use it for short checkpoints and concept-review questions.
- ▶ Your responses help me see which ideas need further clarification.

Registration format

- ▶ **First name:** Student ID
- ▶ **Last name:** Your name



Scan to join Vevox.

Today's Orientation

Homework 5 as the main thread

Main focus this week

- ▶ This session is organized around **Homework 5**.
- ▶ We will use the assignment as the main route into ontology-based evaluation.
- ▶ The goal is not only to understand ontology concepts, but to apply them to an evaluative task.

Please keep Homework 5 open

As we move through the slides, please refer to the Homework 5 specification at the same time.

The lecture is designed to help you understand what the assignment is asking you to build, explain, and submit.

By the end of this session, each group should have a clearer path from *robotics concepts* to an *ontology-based evaluative artifact*.

Which statement best describes the role of a URI in RDF?

- A. It is only a clickable webpage address.
- B. It identifies a resource so that RDF statements can refer to it.
- C. It replaces the need for namespaces.
- D. It is the same thing as an RDF triple.

Review focus

This question reviews the connection among URI, namespace, and RDF statements from the previous session.

This slide is intentionally hidden.

The content is reserved for the in-class session.

Route for This Session

From semantic grounding to Homework 5

Semantic Web Stack → Course Ontology → Common Vocabulary Reuse



OWL DL Axioms → Reasoning over Robotics Concepts → Homework 5

Main point

This session explains how ontology supports *semantic grounding*: making robotics concepts explicit, connected, and reusable enough to support meaningful evaluation.

SHACL-based validation is an important next step, but it is not the main topic today. It will be treated later as an advanced extension.

How to Think about Homework 5

Semantic grounding for robotics evaluation

Choose Robotics Concepts → Ground Their Meanings



Express Ontological Commitments → Explain Evaluation Relevance

Step	What your group should focus on
Choose concepts	Select robotics-related concepts that matter for evaluating a task, capability, dataset, policy, or system.
Ground meanings	Clarify what each concept means, what it refers to, and how it relates to other concepts.
Express commitments	Use ontology structure and OWL DL-style axioms to make important assumptions explicit.
Explain relevance	Show how the grounded concepts help make evaluation more meaningful and inspectable.

Today's Learning Structure

What each part prepares you to do

Part	Function for Homework 5
Semantic Web layer cake	Shows where URI, RDF, RDFS, OWL, and logic belong in the semantic technology stack.
Course ontology example	Shows what an ontology artifact looks like before students build their own.
Common vocabulary reuse	Explains how your ontology can reuse existing terms instead of inventing everything locally.
OWL DL axioms	Introduces how ontologies express class relations, restrictions, and formal assumptions.
Reasoning example	Shows how explicit axioms can support inference, such as whether an object may count as graspable.
Homework deliverables	Clarifies what each group must build, explain, and submit.

Homework 5 Required Repository Structure

Listing 9 (Please read 16.3 carefully)

```
semantic-affordance-grounding/  
|-- README.md  
|-- homework5-report.pdf (or .md)  
|-- ontology/  
|   |-- group-ontology.ttl  
|   |-- inferred-results.ttl  
|   `-- imports/  
|       |-- course-affordance.ttl  
|       `-- ...  
|-- queries/  
|   |-- graspable_objects.rq  
|   `-- task_objects.rq  
|-- results/  
|   |-- graspable_objects_output.txt  
|   `-- screenshots/  
|-- src/  
`-- LICENSE
```

Submission logic

The repository should make the complete semantic grounding workflow reproducible: ontology, imported vocabulary, query, result, documentation, and report.

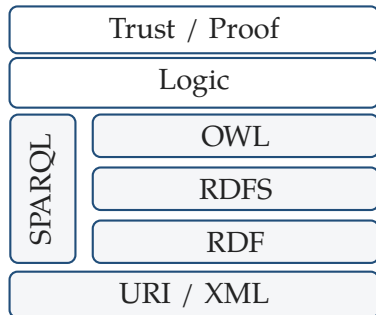
The group-authored ontology is ontology/group-ontology.ttl.

The inferred graph must be saved as ontology/inferred-results.ttl.

The provided course ontology belongs under ontology/imports/.

Semantic Web Layer Cake

Where Homework 5 sits in the stack



How to read the stack

- ▶ Lower layers provide identifiers and graph statements.
- ▶ Middle layers add vocabulary, class structure, and ontology axioms.
- ▶ Upper layers point toward reasoning, proof, and trust.

Homework 5 mainly works in the middle of the stack: URI, RDF, RDFS, OWL, and reasoning over explicitly grounded robotics concepts.

RDF, RDFS, and OWL

Different levels of semantic commitment

Layer	Main function	Example question it can support
RDF	States graph-based facts as triples.	What object was observed in this task?
RDFS	Adds basic vocabulary structure: classes, subclass relations, domains, ranges, labels.	Is every cube also an object?
OWL	Adds richer ontology axioms and formal semantics.	Can an object be inferred as graspable from its defined properties?

For Homework 5, RDF gives the graph, RDFS gives basic vocabulary structure, and OWL gives the axioms needed for reasoning over semantic grounding.

Which layer is mainly responsible for richer axioms that support reasoning?

- A. RDF
- B. RDFS
- C. OWL
- D. URI

Checkpoint focus

This checks whether the three semantic layers are being distinguished by function.

This slide is intentionally hidden.

The content is reserved for the in-class session.

RDF Example

Basic graph statements about an observed object

```
ex:obs_object_01    a                cap:ObservedObject .  
ex:obs_object_01    cap:hasShape     cap:BoxLike .  
ex:obs_object_01    cap:hasWidth     "0.08"^^xsd:decimal .
```

Subject

ex:obs_object_01

Predicate

rdf:type
cap:hasShape
cap:hasWidth

Object

cap:ObservedObject
cap:BoxLike
"0.08"^^xsd:decimal

rdf:type states class membership. Because it is very common, Turtle often abbreviates it as a. RDF provides foundational graph statements that later layers can structure, query, and reason over.

Primary source: World Wide Web Consortium. (2014a). Rdf 1.1 concepts and abstract syntax [W3C Recommendation, 25 February 2014].
<https://www.w3.org/TR/rdf11-concepts/>

RDFS Example

Basic vocabulary structure in the course ontology

```
cap:PhysicalObject    a      rdfs:Class ;  
                      rdfs:label "physical object"@en ;  
                      rdfs:comment "A material object ... assigned a task role."@en .  
  
cap:hasAffordance      a      rdf:Property ;  
                      rdfs:domain cap:PhysicalObject ;  
                      rdfs:range  cap:Affordance ;  
                      rdfs:label  "has affordance"@en ;  
                      rdfs:comment "Relates a physical object ... task context."@en .
```

RDFS gives the ontology its basic vocabulary structure: classes, property domains and ranges, labels, and human-readable comments. In the course ontology, some resources are implemented with OWL vocabulary so that the same terms can later participate in stronger axioms and reasoning patterns. Here they are shown through an RDFS-level reading first.

Primary source: [World Wide Web Consortium. \(2014b\). Rdf schema 1.1 \[W3C Recommendation, 25 February 2014\].](https://www.w3.org/TR/rdf-schema/)
<https://www.w3.org/TR/rdf-schema/>

OWL Levels

Three common ways to understand OWL expressivity

more expressive
less restricted



OWL Lite

simpler classification hierarchies and basic constraints

OWL DL

expressive modeling while preserving decidable reasoning

OWL Full

maximum RDF compatibility and modeling freedom, but without decidability guarantees

For Homework 5, the practical target is **OWL DL-style modeling**: students use ontology axioms to make semantic assumptions explicit while keeping reasoning conceptually controllable.

Primary source: World Wide Web Consortium. (2004). Owl web ontology language reference [W3C Recommendation, 10 February 2004].
<https://www.w3.org/TR/owl-ref/>

DL: Description Logic

A compact notation for class-based reasoning

Basic idea

Description Logic is a family of decidable fragments of First-Order Logic for defining classes through relations, properties, and constraints; it provides the formal basis for OWL DL and Web ontology reasoning.

$\text{cap:GraspableObject} \equiv \text{cap:PhysicalObject} \sqcap \exists \text{cap:hasAffordance.cap:GraspingAffordance}$

$\text{ex:NavigableArea} \equiv \text{ex:Area} \sqcap \exists \text{ex:hasTraversability.ex:Traversable}$
 $\sqcap \neg \exists \text{ex:containsObstacle.ex:Obstacle}$

The first expression is the Homework 5 graspability pattern. The second abstracts a common mobile-robot navigation case: a navigable area is treated as an area that is traversable and does not contain obstacles.

OWL DL Example

The Homework 5 graspability pattern in Turtle

```
cap:GraspableObject
  a owl:Class ;
  owl:equivalentClass [
    a owl:Class ;
    owl:intersectionOf (
      cap:PhysicalObject
      [
        a owl:Restriction ;
        owl:onProperty cap:hasAffordance ;
        owl:someValuesFrom cap:GraspingAffordance
      ]
    )
  ] .
```

This Turtle pattern serializes the DL definition of `cap:GraspableObject`. It says that a graspable object is a physical object that has at least one `cap:hasAffordance` relation to a `cap:GraspingAffordance`. The point is inferability: the reasoner can classify suitable objects as `cap:GraspableObject` from explicit ontology conditions.

In Homework 5, why define `cap:GraspableObject` with OWL conditions?

- A. To manually label every object as graspable.
- B. To let a reasoner infer graspable objects from explicit semantic conditions.
- C. To replace RDF triples with ordinary text.
- D. To avoid defining affordances.

Checkpoint focus

This question checks the core idea of ontology-based semantic grounding: the result should be inferable from explicit commitments.

This slide is intentionally hidden.

The content is reserved for the in-class session.

OWL DL Pattern I

Subclass axiom

`cap:Cup  $\sqsubseteq$  cap:PhysicalObject`

```
cap:Cup  
  rdfs:subClassOf cap:PhysicalObject .
```

This pattern states a class hierarchy: every `cap:Cup` is also a `cap:PhysicalObject`. It is the simplest way to make taxonomy explicit.

Primary source: World Wide Web Consortium. (2012). Owl 2 web ontology language primer [W3C Recommendation, 11 December 2012]. <https://www.w3.org/TR/owl2-primer/>

OWL DL Pattern II

Equivalent class with intersection

$\text{cap:TaskObject} \equiv \text{cap:PhysicalObject} \sqcap \exists \text{cap:hasTaskRole}.\text{cap:ManipulationTarget}$

```
cap:TaskObject
  a owl:Class ;
  owl:equivalentClass [
    a owl:Class ;
    owl:intersectionOf (
      cap:PhysicalObject
      [
        a owl:Restriction ;
        owl:onProperty cap:hasTaskRole ;
        owl:someValuesFrom cap:ManipulationTarget
      ]
    )
  ] .
```

This pattern defines a class by necessary and sufficient conditions. A reasoner can classify individuals into `cap:TaskObject` when the stated conditions are satisfied.

OWL DL Pattern III

Existential restriction

$\text{cap:GraspableObject} \sqsubseteq \exists \text{cap:hasAffordance.cap:GraspingAffordance}$

```
cap:GraspableObject
  rdfs:subClassOf [
    a owl:Restriction ;
    owl:onProperty cap:hasAffordance ;
    owl:someValuesFrom cap:GraspingAffordance
  ] .
```

The existential pattern says that at least one such relation must exist. Here, a graspable object must have some grasping affordance.

Primary source: World Wide Web Consortium. (2012). Owl 2 web ontology language primer [W3C Recommendation, 11 December 2012]. <https://www.w3.org/TR/owl2-primer/>

OWL DL Pattern IV

Universal restriction

$\text{ex:SafeManipulationTask} \sqsubseteq \forall \text{ex:usesTool}.\text{ex:ApprovedTool}$

```
ex:SafeManipulationTask
  rdfs:subClassOf [
    a owl:Restriction ;
    owl:onProperty ex:usesTool ;
    owl:allValuesFrom ex:ApprovedTool
  ] .
```

The universal pattern restricts all values of a property. Here, every tool used in a safe manipulation task must belong to `ex:ApprovedTool`.

Primary source: World Wide Web Consortium. (2012). Owl 2 web ontology language primer [W3C Recommendation, 11 December 2012]. <https://www.w3.org/TR/owl2-primer/>

OWL DL Pattern V

Value restriction

$\text{ex:TabletopPickTask} \sqsubseteq \exists \text{ex:hasWorkSurface}.\{\text{ex:Tabletop}\}$

```
ex:TabletopPickTask
  rdfs:subClassOf [
    a owl:Restriction ;
    owl:onProperty ex:hasWorkSurface ;
    owl:hasValue ex:Tabletop
  ] .
```

The value restriction pattern fixes a property to a specific individual. Here, a tabletop pick task is tied to the particular work surface `ex:Tabletop`.

Primary source: World Wide Web Consortium. (2012). Owl 2 web ontology language primer [W3C Recommendation, 11 December 2012]. <https://www.w3.org/TR/owl2-primer/>

OWL DL Pattern VI

Cardinality restriction

`ex:SingleObjectPickTask  $\sqsubseteq$  (= 1 ex:hasTargetObject.cap:PhysicalObject)`

```
ex:SingleObjectPickTask
  rdfs:subClassOf [
    a owl:Restriction ;
    owl:onProperty ex:hasTargetObject ;
    owl:qualifiedCardinality "1"^^xsd:nonNegativeInteger ;
    owl:onClass cap:PhysicalObject
  ] .
```

The cardinality pattern specifies how many values a property may or must have. Here, a single-object pick task has exactly one target physical object.

Primary source: World Wide Web Consortium. (2012). Owl 2 web ontology language primer [W3C Recommendation, 11 December 2012]. <https://www.w3.org/TR/owl2-primer/>

SPARQL Example

Querying inferred graspable objects

```
PREFIX cap: <https://example.org/aicapstone/ontology#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?obj ?label ?role
WHERE {
    ?obj a cap:GraspableObject .
    OPTIONAL { ?obj cap:hasObjectLabel ?label . }
    OPTIONAL { ?obj cap:hasTaskRole ?role . }
}
ORDER BY ?obj
```

This query should be executed over the **inferred graph**, not only the raw asserted graph. OWL reasoning classifies suitable individuals as `cap:GraspableObject`; SPARQL then retrieves those inferred results with their labels and task roles.

In Homework 5, save the queries as `queries/graspable_objects.rq` and save the outputs under `results/`.

Closing Notes for Homework 5

What to keep in mind

Key points from today's explanation

- ▶ Homework 5 is about **ontology-based semantic grounding**.
- ▶ The repository should make the workflow reproducible.
- ▶ RDF gives graph statements.
- ▶ RDFS gives basic vocabulary structure.
- ▶ OWL DL axioms support reasoning over grounded concepts.
- ▶ SPARQL is used to query inferred results.

What to do next

Please read the full Homework 5 specification carefully. The slides explain the key ideas, but the detailed submission requirements, repository structure, examples, and grading criteria are in the specification.

Due date: same as the Course Project.

Questions are welcome at any time.

Email: ccy@hptp.org

For this homework, you only need to submit the GitHub repository. No presentation or demonstration to the teaching assistants is required. Students who want to go further may join the Individual Study track.